

Questions

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Q1 - 2024 (04 Apr Shift 1)

Let the point, on the line passing through the points $P(1, -2, 3)$ and $Q(5, -4, 7)$, farther from the origin and at distance of 9 units from the point P, be (α, β, γ) . Then $\alpha^2 + \beta^2 + \gamma^2$ is equal to :

(1) 165

(2) 160

(3) 155

(4) 150

Q2 - 2024 (04 Apr Shift 1)

If the shortest distance between the lines $\frac{x+2}{2} = \frac{y+3}{3} = \frac{z-5}{4}$ and $\frac{x-3}{1} = \frac{y-2}{-3} = \frac{z+4}{2}$ is $\frac{38}{3\sqrt{5}}$ k, and

$\int_0^k [x^2] dx = \alpha - \sqrt{\alpha}$, where $[x]$ denotes the greatest integer function, then $6\alpha^3$ is equal to _____

Q3 - 2024 (04 Apr Shift 2)

Let P be the point of intersection of the lines $\frac{x-2}{1} = \frac{y-4}{5} = \frac{z-2}{1}$ and $\frac{x-3}{2} = \frac{y-2}{3} = \frac{z-3}{2}$. Then, the shortest distance of P from the line $4x = 2y = z$ is

(1) $\frac{5\sqrt{14}}{7}$

(2) $\frac{3\sqrt{14}}{7}$

(3) $\frac{\sqrt{14}}{7}$

(4) $\frac{6\sqrt{14}}{7}$

Q4 - 2024 (04 Apr Shift 2)

Consider a line L passing through the points $P(1, 2, 1)$ and $Q(2, 1, -1)$. If the mirror image of the point $A(2, 2, 2)$ in the line L is (α, β, γ) , then $\alpha + \beta + 6\gamma$ is equal to _____

Q5 - 2024 (05 Apr Shift 1)

If the line $\frac{2-x}{3} = \frac{3y-2}{4\lambda+1} = 4-z$ makes a right angle with the line $\frac{x+3}{3\mu} = \frac{1-2y}{6} = \frac{5-z}{7}$, then $4\lambda + 9\mu$ is equal to :

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- (1) 4
- (2) 13
- (3) 5
- (4) 6

Q6 - 2024 (05 Apr Shift 1)

Let d be the distance of the point of intersection of the lines $\frac{x+6}{3} = \frac{y}{2} = \frac{z+1}{1}$ and $\frac{x-7}{4} = \frac{y-9}{3} = \frac{z-4}{2}$ from the point $(7, 8, 9)$. Then $d^2 + 6$ is equal to :

- (1) 69
- (2) 78
- (3) 72
- (4) 75

Q7 - 2024 (05 Apr Shift 2)

Let (α, β, γ) be the image of the point $(8, 5, 7)$ in the line $\frac{x-1}{2} = \frac{y+1}{3} = \frac{z-2}{5}$. Then $\alpha + \beta + \gamma$ is equal to :

- (1) 16
- (2) 20
- (3) 14
- (4) 18

Q8 - 2024 (05 Apr Shift 2)

Let the point $(-1, \alpha, \beta)$ lie on the line of the shortest distance between the lines $\frac{x+2}{-3} = \frac{y-2}{4} = \frac{z-5}{2}$ and $\frac{x+2}{-1} = \frac{y+6}{2} = \frac{z-1}{0}$. Then $(\alpha - \beta)^2$ is equal to _____

Q9 - 2024 (06 Apr Shift 1)

The shortest distance between the lines $\frac{x-3}{2} = \frac{y+15}{-7} = \frac{z-9}{5}$ and $\frac{x+1}{2} = \frac{y-1}{1} = \frac{z-9}{-3}$ is

- (1) $8\sqrt{3}$

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(2) $4\sqrt{3}$

(3) $5\sqrt{3}$

(4) $6\sqrt{3}$

Q10 - 2024 (06 Apr Shift 1)

Let P be the point $(10, -2, -1)$ and Q be the foot of the perpendicular drawn from the point $R(1, 7, 6)$ on the line passing through the points $(2, -5, 11)$ and $(-6, 7, -5)$. Then the length of the line segment PQ is equal to

Q11 - 2024 (06 Apr Shift 2)

Let $P(\alpha, \beta, \gamma)$ be the image of the point $Q(3, -3, 1)$ in the line $\frac{x-0}{1} = \frac{y-3}{1} = \frac{z-1}{-1}$ and R be the point $(2, 5, -1)$. If the area of the triangle PQR is λ and $\lambda^2 = 14K$, then K is equal to :

(1) 36

(2) 81

(3) 72

(4) 18

Q12 - 2024 (06 Apr Shift 2)

If the shortest distance between the lines $\frac{x-\lambda}{3} = \frac{y-2}{-1} = \frac{z-1}{1}$ and $\frac{x+2}{-3} = \frac{y+5}{2} = \frac{z-4}{4}$ is $\frac{44}{\sqrt{30}}$, then the largest possible value of $|\lambda|$ is equal to

Q13 - 2024 (08 Apr Shift 1)

If the shortest distance between the lines

$$L_1 : \vec{r} = (2 + \lambda)\hat{i} + (1 - 3\lambda)\hat{j} + (3 + 4\lambda)\hat{k}, \quad \lambda \in \mathbb{R}$$

$$L_2 : \vec{r} = 2(1 + \mu)\hat{i} + 3(1 + \mu)\hat{j} + (5 + \mu)\hat{k}, \quad \mu \in \mathbb{R}$$

is $\frac{m}{\sqrt{n}}$, where $\gcd(m, n) = 1$, then the value of $m + n$ equals

(1) 390

(2) 384

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(3) 377

(4) 387

Q14 - 2024 (08 Apr Shift 1)

Let $P(x, y, z)$ be a point in the first octant, whose projection in the xy -plane is the point Q . Let $OP = \gamma$; the angle between OQ and the positive x -axis be θ ; and the angle between OP and the positive z -axis be ϕ , where O is the origin. Then the distance of P from the x -axis is

(1) $\gamma\sqrt{1 - \sin^2 \phi \cos^2 \theta}$

(2) $\gamma\sqrt{1 - \sin^2 \theta \cos^2 \phi}$

(3) $\gamma\sqrt{1 + \cos^2 \phi \sin^2 \theta}$

(4) $\gamma\sqrt{1 + \cos^2 \theta \sin^2 \phi}$

Q15 - 2024 (08 Apr Shift 2)

If the shortest distance between the lines $\frac{x-\lambda}{2} = \frac{y-4}{3} = \frac{z-3}{4}$ and $\frac{x-2}{4} = \frac{y-4}{6} = \frac{z-7}{8}$ is $\frac{13}{\sqrt{29}}$, then a value of λ is :

(1) -1

(2) $-\frac{13}{25}$

(3) $\frac{13}{25}$

(4) 1

Q16 - 2024 (08 Apr Shift 2)

Let $P(\alpha, \beta, \gamma)$ be the image of the point $Q(1, 6, 4)$ in the line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$. Then $2\alpha + \beta + \gamma$ is equal to _____

Q17 - 2024 (09 Apr Shift 1)

Let the line L intersect the lines $x - 2 = -y = z - 1$, $2(x + 1) = 2(y - 1) = z + 1$ and be parallel to the line $\frac{x-2}{3} = \frac{y-1}{1} = \frac{z-2}{2}$. Then which of the following points lies on L ?

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- (1) $\left(-\frac{1}{3}, 1, -1\right)$
- (2) $\left(-\frac{1}{3}, -1, 1\right)$
- (3) $\left(-\frac{1}{3}, 1, 1\right)$
- (4) $\left(-\frac{1}{3}, -1, -1\right)$

Q18 - 2024 (09 Apr Shift 1)

The shortest distance between the lines $\frac{x-3}{4} = \frac{y+7}{-11} = \frac{z-1}{5}$ and $\frac{x-5}{3} = \frac{y-9}{-6} = \frac{z+2}{1}$ is:

- (1) $\frac{178}{\sqrt{563}}$
- (2) $\frac{187}{\sqrt{563}}$
- (3) $\frac{185}{\sqrt{563}}$
- (4) $\frac{179}{\sqrt{563}}$

Q19 - 2024 (09 Apr Shift 2)

Consider the line L passing through the points $(1, 2, 3)$ and $(2, 3, 5)$. The distance of the point $\left(\frac{11}{3}, \frac{11}{3}, \frac{19}{3}\right)$ from the line L along the line $\frac{3x-11}{2} = \frac{3y-11}{1} = \frac{3z-19}{2}$ is equal to

- (1) 6
- (2) 5
- (3) 4
- (4) 3

Q20 - 2024 (09 Apr Shift 2)

The square of the distance of the image of the point $(6, 1, 5)$ in the line $\frac{x-1}{3} = \frac{y}{2} = \frac{z-2}{4}$, from the origin is

- _____
- _____
- _____
- _____
- _____

Questions

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Answer Key

Q1 (3)	Q2 (48)	Q3 (2)	Q4 (6)		
Q5 (4)	Q6 (4)	Q7 (3)	Q8 (25)		
Q9 (2)	Q10 (13)	Q11 (2)	Q12 (43)		
Q13 (4)	Q14 (1)	Q15 (4)	Q16 (11)		
Q17 (1)	Q18 (2)	Q19 (4)	Q20 (62)		

Solutions

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Q1

PQ line

$$\frac{x-1}{4} = \frac{y+2}{-2} = \frac{z-3}{4}$$

$$\text{pt}(4t+1, -2t-2, 4t+3)$$

$$\text{distance}^2 = 16t^2 + 4t^2 + 16t^2 = 81$$

$$t = \pm \frac{3}{2}$$

$$\text{pt}(7, -5, 9)$$

$$\alpha^2 + \beta^2 + \gamma^2 = 155$$

option (1)

Q2

$$\frac{38}{3\sqrt{5}}\hat{k} = \frac{(5\hat{i} + 5\hat{j} - 9\hat{k})}{\sqrt{5}} \cdot \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 3 & 4 \\ 1 & -3 & 2 \end{vmatrix}$$

$$\frac{38}{3\sqrt{5}}\hat{k} = \frac{19}{\sqrt{5}}$$

$$k = \frac{19}{\sqrt{5}}$$

$$k = \frac{3}{2}$$

$$\int_0^{3/2} [x^2] = \int_0^1 0 + \int_1^{\sqrt{2}} 1 + \int_{\sqrt{2}}^{3/2} 2$$

$$= \sqrt{2} - 1 + 2\left(\frac{3}{2} - \sqrt{2}\right)$$

$$= 2 - \sqrt{2}$$

$$\alpha = 2$$

$$\Rightarrow 6\alpha^3 = 48$$

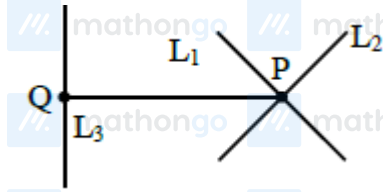
Q3

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Solutions

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$$L_1 \equiv \frac{x-2}{1} = \frac{y-4}{5} = \frac{z-2}{1} = \lambda$$

$$P(\lambda + 2, 5\lambda + 4, \lambda + 2)$$

$$L_2 \equiv \frac{x-3}{2} = \frac{y-2}{3} = \frac{z-3}{2}$$

$$P(2\mu + 3, 3\mu + 2, 2\mu + 3)$$

$$\lambda + 2 = 2\mu + 3 \quad 3\mu + 2 = 5\lambda + 4$$

$$\lambda = 2\mu + 1 \quad 3\mu = 5\lambda + 2$$

$$3\mu = 5(2\mu + 1) + 2$$

$$3\mu = 10\mu + 7$$

$$\mu = -1 \quad \lambda = -1$$

Both satisfies (P)

$$P(1, -1, 1)$$

$$L_3 \equiv \frac{x}{1/4} = \frac{y}{1/2} = \frac{z}{1}$$

$$L_3 \equiv \frac{x}{1} = \frac{y}{2} = \frac{z}{4} = k$$

Coordinates of Q(k, 2k, 4k)

$$\text{DR's PQ} = \langle k-1, 2k+1, 4k-1 \rangle$$

$$PQ \perp \text{to } L_3$$

$$(k-1) + 2(2k+1) + 4(4k-1) = 0$$

$$k-1 + 4k+2 + 16k-4 = 0$$

$$k = \frac{1}{7}$$

$$Q\left(\frac{1}{7}, \frac{2}{7}, \frac{4}{7}\right)$$

$$PQ = \sqrt{\left(1 - \frac{1}{7}\right)^2 + \left(-1 - \frac{2}{7}\right)^2 + \left(1 - \frac{4}{7}\right)^2}$$

$$= \sqrt{\frac{36}{49} + \frac{81}{49} + \frac{9}{49}} = \frac{\sqrt{126}}{7}$$

$$PQ = \frac{3\sqrt{14}}{7}$$

Q4

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Solutions

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$$\text{DR's of Line } L \equiv -1 : 1 : 2$$

$$\text{DR's of } AB \equiv \alpha - 2 : \beta - 2 : \gamma - 2$$

$$AB \perp_{\text{ar}} L \Rightarrow 2 - \alpha + \beta - 2 + 2\gamma - 4 = 0$$

$$2\gamma + \beta - \alpha = 4$$

Let C is mid-point of AB

$$C \left(\frac{\alpha + 2}{2}, \frac{\beta + 2}{2}, \frac{\gamma + 2}{2} \right)$$

$$\text{DR's of } PC = \frac{\alpha}{2} : \frac{\beta - 2}{2} : \frac{\gamma}{2}$$

$$\text{line } L \perp PC \Rightarrow \frac{-\alpha}{2} = \frac{\beta - 2}{2} = \frac{\gamma}{4} = K \text{ (let)}$$

$$\alpha = -2K$$

$$\beta = 2K + 2$$

$$\gamma = 4K$$

$$\text{use in (1)} \Rightarrow K = \frac{1}{6}$$

$$\text{value of } \alpha + \beta + 6\gamma = 24K + 2 = 6$$

Q5

$$\frac{2-x}{3} = \frac{3y-2}{4\lambda+1} = 4 - z \dots (1)$$

$$\frac{x-2}{(-3)} = \frac{y-\frac{2}{3}}{\left(\frac{4\lambda+1}{3}\right)} = \frac{z-4}{(-1)}$$

$$\frac{x+3}{3\mu} = \frac{1-2y}{6} = \frac{5-z}{7} \dots (2)$$

$$\frac{x+3}{3\mu} = \frac{y-\frac{1}{2}}{(-3)} = \frac{z-5}{(-7)}$$

$$\text{Right angle} \Rightarrow (-3)(3\mu) + \left(\frac{4\lambda+1}{3}\right)(-3) + (-1)(-7) = 0$$

$$-9\mu - 4\lambda - 1 + 7 = 0$$

$$4\lambda + 9\mu = 6$$

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Solutions

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Q6

$$\frac{x+6}{3} = \frac{y}{2} = \frac{z+1}{1} = \lambda \dots (1)$$

$$x = 3\lambda - 6, y = 2\lambda, z = \lambda - 1$$

$$\frac{x-7}{4} = \frac{y-9}{3} = \frac{z-4}{2} = \mu \dots (2)$$

$$x = 4\mu + 7, y = 3\mu + 9, z = 2\mu + 4$$

$$3\lambda - 6 = 4\mu + 7 \Rightarrow 3\lambda - 4\mu = 13 \dots (3) \times 2$$

$$2\lambda = 3\mu + 9 \Rightarrow 2\lambda - 3\mu = 9 \dots (4) \times 3$$

$$6\lambda - 8\mu = 26$$

$$6\lambda - 9\mu = 27$$

$$\begin{array}{r} - \quad + \quad - \\ \hline \mu = -1 \end{array}$$

$$\Rightarrow 3\lambda - 4(-1) = 13$$

$$3\lambda = 9$$

$$\lambda = 3$$

int. point (3, 6, 2); (7, 8, 9)

$$d^2 = 16 + 4 + 49 = 69$$

$$\text{Ans. } d^2 + 6 = 69 + 6 = 75$$

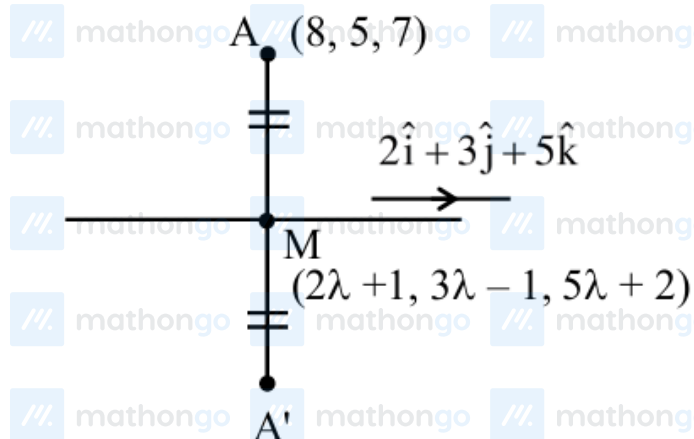
Q7

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Solutions

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$$\overrightarrow{AM} \cdot (2\hat{i} + 3\hat{j} + 5\hat{k}) = 0$$

$$(2\lambda - 7)(2) + (3\lambda - 6)(3) + (5\lambda - 5)(5) = 0$$

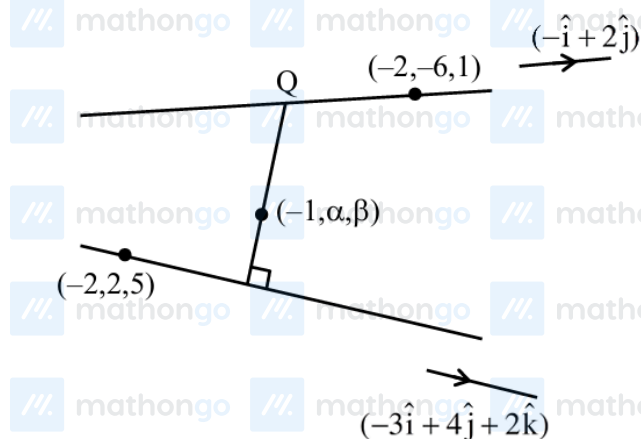
$$38\lambda = 57$$

$$\lambda = \frac{3}{2}$$

$$M\left(4, \frac{7}{2}, \frac{19}{2}\right)$$

$$A'(0, 2, 12)$$

Q8



$$P(-3\lambda - 2, 4\lambda + 2, 2\lambda + 5)$$

$$Q(-\mu - 2, 2\mu - 6, 1)$$

$$\text{DRS of } PQ = (3\lambda - \mu, 2\mu - 4\lambda - 8, -2\lambda - 4)$$

$$\text{DRS of } PQ = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 2 & 0 \\ -3 & 4 & 2 \end{vmatrix}$$

$$= (4\hat{i} + 2\hat{j} + 2\hat{k})$$

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Solutions

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OR

$$(2, 1, 1)$$

$$\frac{3\lambda - \mu}{2} = \frac{2\mu - 4\lambda - 8}{1} = \frac{-2\lambda - 4}{1}$$

$$\Rightarrow \mu = \lambda + 2 \text{ and } 7\lambda = \mu - 8$$

$$\lambda = -1 \quad \mu = 1$$

$$Q : (-3, -4, 1)$$

$$L_{PQ} = \frac{x+3}{2} = \frac{y+4}{1} = \frac{z-1}{1}$$

$$(-1, \alpha, \beta) \Rightarrow 1 = \frac{\alpha+4}{1} = \frac{\beta-1}{1}$$

$$\Rightarrow \alpha = -3, \beta = 2$$

$$(\alpha - \beta)^2 = 25$$

Q9

$$\frac{x-3}{2} = \frac{y+15}{-7} = \frac{z-9}{5} \text{ \& } \frac{x+1}{2} = \frac{y-1}{1} = \frac{z-9}{-3}$$

$$\text{S.D} = \frac{|(\bar{a}_2 \cdot \bar{a}_1) \cdot (\bar{b}_1 \cdot \bar{b}_2)|}{|\bar{b}_1 \times \bar{b}_2|}$$

$$a_1 = 3, -15, 9$$

$$b_1 = 2, -7, 5$$

$$a_2 = -1, 1, 9$$

$$b_2 = 2, 1, -3$$

$$a_2 - a_1 = -4, 16, 0$$

$$\bar{b}_1 \times \bar{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -7 & 5 \\ 2 & 1 & -3 \end{vmatrix} = \hat{i}(16) - \hat{j}(-16) + \hat{k}(16)$$

$$16(\hat{i} + \hat{j} + \hat{k})$$

$$|\bar{b}_1 \times \bar{b}_2| = 16\sqrt{3}$$

$$\therefore (\bar{a}_2 - \bar{a}_1) \cdot (\bar{b}_1 - \bar{b}_2) = 16[-4 + 16] = (16)(12)$$

$$\text{S.D.} = \frac{(16)(12)}{16\sqrt{3}} = 4\sqrt{3}$$

Q10

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Solutions

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$$\text{Line : } \frac{x+6}{-8} = \frac{y-7}{12} = \frac{z+5}{-16}$$

$$\frac{x+6}{2} = \frac{y-7}{-3} = \frac{z+5}{4} = \lambda$$

$$Q(2\lambda - 6, 7 - 3\lambda, 4\lambda - 5)$$

$$QR(2\lambda - 7, -3\lambda, 4\lambda - 11)$$

$$\overline{QR} \cdot \text{dr's of line} = 0$$

$$4\lambda - 14 + 9\lambda + 16\lambda - 44 = 0$$

$$29\lambda = 58 \Rightarrow \lambda = 2$$

$$Q(-2, 1, 3)$$

$$PQ = \sqrt{144 + 9 + 16} = \sqrt{169} = 13$$

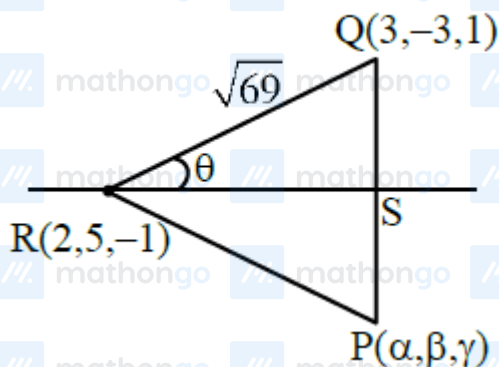
Q11

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Solutions

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$$RQ = \sqrt{1 + 64 + 4} = \sqrt{69}$$

$$\vec{RQ} = \hat{i} - 8\hat{j} + 2\hat{k}$$

$$\vec{RS} = \hat{i} + \hat{j} - \hat{k}$$

$$\cos \theta = \frac{\vec{RQ} \cdot \vec{RS}}{|\vec{RQ}| |\vec{RS}|} = \frac{|1 - 8 - 2|}{\sqrt{69} \sqrt{3}} = \frac{9}{3\sqrt{23}}$$

$$\cos \theta = \frac{3}{\sqrt{23}} = \frac{RS}{RQ} = \frac{RS}{\sqrt{69}}$$

$$RS = 3\sqrt{3}$$

$$\sin \theta = \frac{\sqrt{14}}{\sqrt{23}} = \frac{QS}{\sqrt{69}}$$

$$QS = \sqrt{42}$$

$$\text{area} = \frac{1}{2} \cdot 2QS \cdot RS = \sqrt{42} \cdot 3\sqrt{3}$$

$$\lambda = 9\sqrt{14}$$

$$\lambda^2 = 81 \cdot 14 = 14k$$

$$k = 81$$

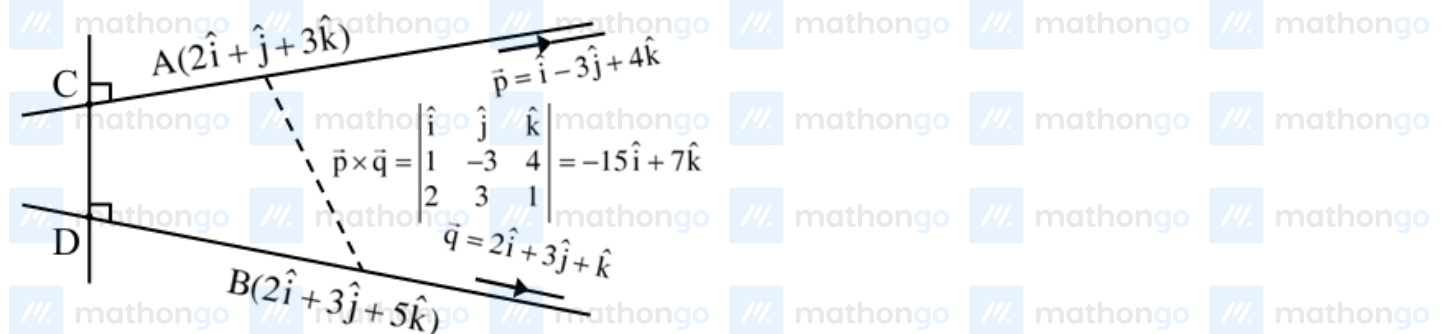
Q12

Solutions

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$$\begin{aligned}\vec{a}_1 &= \lambda \hat{i} + 2\hat{j} + \hat{k} \\ \vec{a}_2 &= -2\hat{i} - 5\hat{j} + 4\hat{k} \\ \vec{p} &= 3\hat{i} - \hat{j} + \hat{k} \\ \vec{q} &= -3\hat{i} + 2\hat{j} + 4\hat{k} \\ (\lambda + 2)\hat{i} + 7\hat{j} - 3\hat{k} &= \vec{a}_1 - \vec{a}_2 \\ \vec{p} \times \vec{q} &= -6\hat{i} - 15\hat{j} + 3\hat{k} \\ 44 &= \frac{|\vec{p} \times \vec{q}|}{\sqrt{30}} = \frac{\sqrt{(-6)^2 + (-15)^2 + 3^2}}{\sqrt{30}} \\ 44 &= \frac{|6\lambda + 126|}{3\sqrt{30}} \\ 132 &= |6\lambda + 126| \\ \lambda &= 1, \lambda = -43 \\ |\lambda| &= 43\end{aligned}$$

Q13



$$\begin{aligned}\text{Shortest distance (CD)} &= \frac{|\vec{AB} \cdot \vec{p} \times \vec{q}|}{|\vec{p} \times \vec{q}|} \\ &= \frac{|(0\hat{i} + 2\hat{j} + 2\hat{k}) \cdot (-15\hat{i} + 7\hat{j} + 9\hat{k})|}{\sqrt{355}} \\ &= \frac{0 + 14 + 18}{\sqrt{355}} = \frac{32}{\sqrt{355}} \\ \therefore m + n &= 32 + 355 = 387\end{aligned}$$

Q14

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Solutions

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$$P(x, y, z), Q(x, y, 0); x^2 + y^2 + z^2 = \gamma^2$$

$$\overrightarrow{OQ} = x\hat{i} + y\hat{j}$$

$$\cos \theta = \frac{x}{\sqrt{x^2 + y^2}}$$

$$\cos \phi = \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$

$$\Rightarrow \sin^2 \phi = \frac{x^2 + y^2}{x^2 + y^2 + z^2}$$

$$\text{distance of P from x-axis} = \sqrt{y^2 + z^2}$$

$$\Rightarrow \sqrt{\gamma^2 - x^2} \Rightarrow \gamma \sqrt{1 - \frac{x^2}{\gamma^2}}$$

$$= \gamma \sqrt{1 - \cos^2 \theta \sin^2 \phi}$$

Q15

$$\left. \begin{aligned} \vec{r}_1 &= (\lambda\hat{i} + 4\hat{j} + 3\hat{k}) + \alpha(2\hat{i} + 3\hat{j} + 4\hat{k}) \\ \vec{r}_2 &= (2\hat{i} + 4\hat{j} + 7\hat{k}) + \beta(2\hat{i} + 3\hat{j} + 4\hat{k}) \end{aligned} \right\} \begin{aligned} \vec{b} &= 2\hat{i} + 3\hat{j} + 4\hat{k} \\ \vec{a}_2 &= \lambda\hat{i} + 4\hat{j} + 3\hat{k} \\ \vec{a}_2 &= 2\hat{i} + 4\hat{j} + 7\hat{k} \end{aligned}$$

$$\text{Shortest dist.} = \frac{|\vec{b} \times (\vec{a}_2 - \vec{a}_1)|}{|\vec{b}|} = \frac{13}{\sqrt{29}}$$

$$\left| \frac{(2\hat{i} + 3\hat{j} + 4\hat{k}) \times ((2 - \lambda)\hat{i} + 4\hat{k})}{\sqrt{29}} \right| = \frac{13}{\sqrt{29}}$$

$$|-8\hat{j} - 3(2 - \lambda)\hat{k} + 12\hat{i} + 4(2 - \lambda)\hat{j}| = 13$$

$$|12\hat{i} - 4\lambda\hat{j} + (3\lambda - 6)\hat{k}| = 13$$

$$144 + 16\lambda^2 + (3\lambda - 6)^2 = 169$$

$$16\lambda^2 + (3\lambda - 6)^2 = 25 = \lambda \Rightarrow \lambda = 1$$

Q16

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Solutions

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Q(1, 6, 4)

A $\left(\frac{17}{14}, \frac{48}{14}, \frac{79}{14}\right)$

$$\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$$

P(α, β, γ)

$$\vec{b} = \hat{i} + 2\hat{j} + 3\hat{k}$$

$$A(t, 2t+1, 3t+2)$$

$$\vec{QA} = (t-1)\hat{i} + (2t-5)\hat{j} + (3t-2)\hat{k}$$

$$\vec{QA} \cdot \vec{b} = 0$$

$$(t-1) + 2(2t-5) + 3(3t-2) = 0$$

$$14t = 17$$

$$\alpha = \frac{20}{14}, \beta = \frac{12}{14}, \gamma = \frac{102}{14}$$

$$2\alpha + \beta + \gamma = \frac{154}{14} = 11$$

Q17

L₁

dr's of line(3, 1, 2)

M(2+ λ , - λ , 1+ λ)L₂N $\left(-1+\frac{\mu}{2}, 1+\frac{\mu}{2}, -1+\mu\right)$

$$L_1 : \frac{x-2}{1} = \frac{y}{-1} = \frac{z-1}{1} = \lambda$$

$$L_2 : \frac{x+1}{\frac{1}{2}} = \frac{y-1}{\frac{1}{2}} = \frac{z+1}{1} = \mu$$

dr of line MN will be

$$\langle 3 + \lambda - \frac{\mu}{2}, -1 - \lambda - \frac{\mu}{2}, 2 + \lambda - \mu \rangle \text{ \& it will be proportional to } \langle 3, 1, 2 \rangle$$

Solutions

MathonGo

$$\therefore \frac{3+\lambda-\frac{\mu}{2}}{3} = \frac{-1-\lambda-\frac{\mu}{2}}{1} = \frac{2+\lambda-\mu}{2}$$

$$4\lambda + \mu = -6 \quad 4 + 3\lambda = 0$$

$$\Rightarrow \lambda = -\frac{4}{3} \text{ \& } \mu = -\frac{2}{3}$$

$\therefore$ Coordinate of M will be $\left(\frac{2}{3}, \frac{4}{3}, -\frac{1}{3}\right)$ and equation of required line will be.

$$\frac{x-\frac{2}{3}}{3} = \frac{y-\frac{4}{3}}{1} = \frac{z+\frac{1}{3}}{2} = k$$

So any point on this line will be

$$\left(\frac{2}{3} + 3k, \frac{4}{3} + k, -\frac{1}{3} + 2k\right)$$

$$\therefore \frac{2}{3} + 3k = -\frac{1}{3} \Rightarrow k = -\frac{1}{3}$$

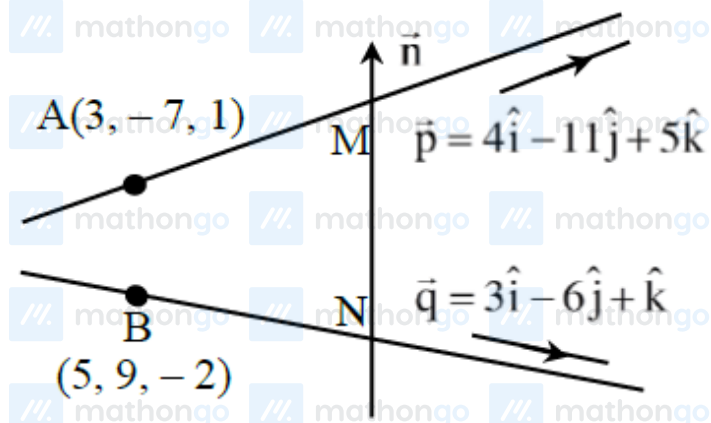
$\therefore$ Point lie on the line for

$$k = -\frac{1}{3} \text{ is } \left(-\frac{1}{3}, 1, -1\right)$$

Q18

Solutions

MathonGo



$$\vec{n} = \vec{p} \times \vec{q}$$

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 4 & -11 & 5 \\ 3 & -6 & 1 \end{vmatrix} = 19\hat{i} + 11\hat{j} + 9\hat{k}$$

S.d. = projection of $\vec{AB}$ on $\vec{n}$

$$= \frac{|\vec{AB} \cdot \vec{n}|}{|\vec{n}|} = \frac{|(2\hat{i} + 16\hat{j} - 3\hat{k}) \cdot (19\hat{i} + 11\hat{j} + 9\hat{k})|}{\sqrt{361 + 121 + 81}}$$

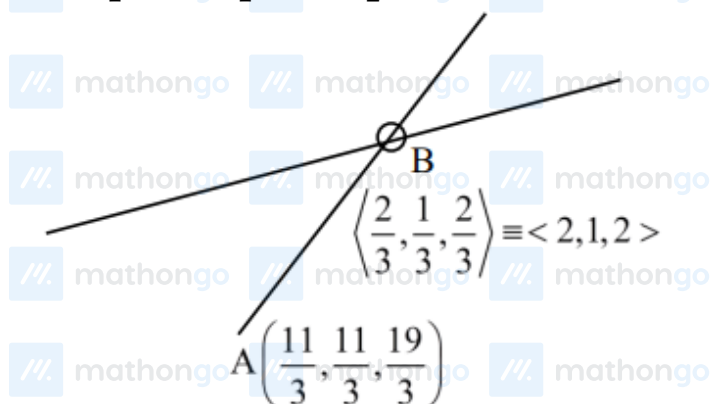
$$= \frac{38 + 176 - 27}{\sqrt{563}}$$

$$\text{S.d.} = \frac{187}{\sqrt{563}}$$

Q19

$$\frac{x-1}{2-1} = \frac{y-2}{3-2} = \frac{z-3}{5-3}$$

$$\Rightarrow \frac{x-1}{1} = \frac{y-2}{1} = \frac{z-3}{2} = \lambda$$



Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

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Solutions

MathonGo

$$B(1 + \lambda, 2 + \lambda, 3 + 2\lambda)$$

$$\text{D.R. of AB} = \left\langle \frac{3\lambda - 8}{3}, \frac{3\lambda - 5}{3}, \frac{6\lambda - 10}{3} \right\rangle$$

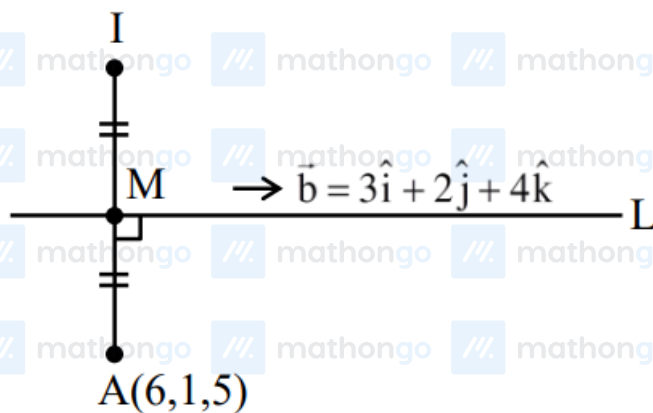
$$B\left(\frac{5}{3}, \frac{8}{3}, \frac{13}{3}\right) \frac{3\lambda - 8}{3\lambda - 5} = \frac{2}{1} \Rightarrow 3\lambda - 8 = 6\lambda - 10$$

$$3\lambda = 2$$

$$\lambda = \frac{2}{3}$$

$$AB = \frac{\sqrt{36 + 9 + 36}}{3} = \frac{9}{3} = 3$$

Q20



$$\text{Let } M(3\lambda + 1, 2\lambda, 4\lambda + 2)$$

$$\vec{AM} \cdot \vec{b} = 0$$

$$\Rightarrow 9\lambda - 15 + 4\lambda - 2 + 16\lambda - 12 = 0$$

$$\Rightarrow 29\lambda = 29$$

$$\Rightarrow \lambda = 1$$

$$M(4, 2, 6), I = (2, 3, 7)$$

$$\text{Required Distance} = \sqrt{4 + 9 + 49} = \sqrt{62}$$

Ans. 62